Parameters extraction of photovoltaic module at reference and real conditions

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Abstract - This paper deals with design procedure, simulation and experimentation of extraction parameters of photovoltaic module based an equivalent electrical circuit of the single diode said five model parameters or a complete model. Using an analytic method based on some approximation for extracting the five photovoltaic module parameters of the single diode lumped circuit model. These parameters are usually the photocurrent, the saturation current, the ideality factor, the series resistance and the shunt resistance. The behaviour of the photovoltaic module is described by a current-voltage characteristic I-V in which its shape is greatly depended on the values of these parameters beside their dependence on solar irradiance and module temperature. Therefore, parameters extraction has been carried out mainly based on analytical method and experimental measurement on a PV module Isofoton12-106. The predicted current-voltage curves are compared with experimental data provided by the Centre of Development of Renewable Energies, Bouzaréah, Algeria, to conclude on the validity of the model and the followed procedure.

Index Terms-- analytical method, Cell, diode model, extraction, parameters, photovoltaic, prediction, real conditions.

I. INTRODUCTION

The behaviour of photovoltaic cell/module is usually described by its current-voltage characteristic, in which their shapes are greatly dependent on the values of the five parameters Iph, I0, A, Rs, Rsh, beside their dependence on solar irradiance and module temperature. The determination of these parameters at any solar irradiance and module temperature conditions makes easy to apply the model for predicting the energy production of photovoltaic module. Therefore, parameters extraction is carried out mainly based on analytical method and experimental measurement on a PV module Isofoton12-106. The extraction of the five parameters allows the comparison between predicted and measured characteristics to be made. It provides theoretical studies of photovoltaics and modeling techniques using equivalent electric circuits

II. PHOTOVOLTAIC CELL MODEL

Photovoltaic cell models have long been a source for the description of photovoltaic cell behaviour. The most common model used to predict energy production in photovoltaic cell modeling is the single diode lumped circuit model [1], shown in Fig. 1.

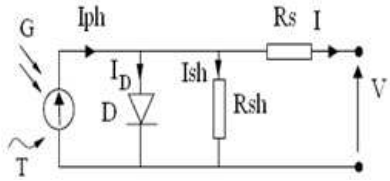

Fig. 1 Equivalent Electrical circuit of the single diode model.

This model has a current source Iph, which depends on solar radiation and cell temperature, a diode in parallel which the inverse saturation intensity I0 depends on temperature, a series resistance Rs, which represents the effect of internal resistance and cell contacts, and a shunt resistance Rsh in parallel with the diode, which represents the current flowing through the crystal.

The current–voltage characteristic of a photovoltaic cell is described by

$$I = I_{ph} - I_0\left(\exp\left(\frac{V + R_s I}{A}\right) - 1\right) - \frac{V + R_s I}{R_{sh}} \qquad (1)$$

Where:
I_{ph}: Photocurrent (A).
I_0: Saturation current (A).
A=nVt (V): modified ideal factor.
n: Diode ideality factor.
k: Boltzmann constante ($k=1.38\ 10^{-23}\ J\,K^{-1}$).
q: Electron charge ($q=1.602\ 10^{-19}$ C).
T: Cell temperature (°K).
Vt: Thermique voltage.Vt=kT/q (V)
R_s: Series resistance (Ω).
R_{sh}: Shunt resistance (Ω).

The non linear and implicit equation given by (1) can be solved using the Newton Raphson iterative method [2].

The five parameters appearing in equation (1) depend on the incident solar irradiance, the cell temperature, and on their reference values. These reference values are generally provided by manufacturers of photovoltaic generators for specified operating condition such as STC (standard testing conditions) for which the irradiance 1000W/m² and the cell temperature is 25°C. The aim of this work is to calculate the

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five parameters at reference conditions and at any other condition of solar irradiance and cell temperature to provide valuable model for photovoltaic energy prediction.

III. 3DETERMINATION OF REFERENCE PARAMETERS

At reference conditions, equation (1) can be written as:

$$I = I_{phref} - I_{0ref}\left(\exp\left(\frac{V + R_{sref} I}{A_{ref}}\right) - 1\right) - \frac{V + R_{sref} I}{R_{shref}} \quad (2)$$

To find the reference parameters Iphref, I0ref, Aref, Rsref, and Rshref we use the following points [3]:

At short circuit point: I=ISC and V=0

$$I_{scref} - I_{phref} + I_{0ref}\left\{\exp\left(\frac{I_{scref}R_{sref}}{A_{ref}}\right) - 1\right\} + \frac{I_{scref}R_{sref}}{R_{shref}} = 0 \quad (3)$$

At open circuit point: I=0 and V=V0C

$$I_{phref} - I_{0ref}\left\{\exp\left(\frac{V_{ocref}}{A_{ref}}\right) - 1\right\} - \frac{V_{ocref}}{R_{shref}} = 0 \quad (4)$$

At maximum power point: V=Vm and I=Im

$$I_{mref} = I_{phref} - I_{0ref}\left\{\exp\left(\frac{V_{mref} + I_{mref}R_{s,ref}}{A_{ref}}\right) - 1\right\} - \frac{V_{mref} + I_{mref}R_{sref}}{R_{shref}} \quad (5)$$

The derivation of equation (1) at V=V0C:

$$\frac{dV}{dI}\Big|_{V_{oc}} = -R_{S0} \quad (6)$$

The derivation at I =ISC:

$$\frac{dV}{dI}\Big|_{I_{sc}} = -R_{sh0} \quad (7)$$

Where Rs0 is the slop at open circuit point, and Rsh0 is the slop at short circuit point. In this work, Rs0 and Rsh0 are taken from manufacturer's data.

From equation (4), we derive the expression of the photocurrent Iphref and we replace it in equation (3) and (5), after some mathematics manipulations we obtain then the following system equations with four parameters:

$$I_{0ref}\left\{\exp\frac{V_{ocref}}{A_{ref}} - \exp\frac{I_{scref}R_{sref}}{A_{ref}}\right\} - I_{scref}\left(1 + \frac{R_{sref}}{R_{shref}}\right) + \frac{V_{ocref}}{R_{shref}} = 0$$

$$I_{mref} - I_{0ref}\left\{\exp\left(\frac{V_{ocref}}{A_{ref}}\right) - \exp\left(\frac{V_{mref} + I_{mref}R_{sref}}{A_{ref}}\right)\right\} + \frac{V_{mref} + I_{mref}R_{sref}}{R_{shref}} - \frac{V_{ocref}}{R_{shref}} = 0 \quad (8)$$

$$\left(R_{so} - R_{sref}\right)\left\{\frac{1}{R_{shref}} + \frac{I_{0ref}}{A_{ref}}\exp\left(\frac{V_{ocref}}{A_{ref}}\right)\right\} - 1 = 0$$

$$\frac{1}{R_{shref}} - \frac{1}{R_{sho} - R_{sref}} + \frac{I_{0ref}}{A_{ref}}\exp\frac{I_{scref}R_{sref}}{A_{ref}} = 0$$

Where Iscref, Vocref, Imref, and Vmref are usually given by manufacturers and in laboratory conditions, we propose calculating these parameters at reference conditions using real measures and translation equations as in [4].

$$I_{STC} = I_{mes}\left(\frac{G_{STC}}{G_{mes}}\right) + \alpha\left(T - T_{STC}\right) \quad (9)$$

$$V_{STC} = V_{mes} - \beta\left(T_{STC} - T\right) - R_{smes}\left(I_{mes} - I_{STC}\right) + V_t \ln\left(\frac{G_{STC}}{G_{mes}}\right) \quad (10)$$

With ISTC is the module current at STC (A), GSTC is the reference irradiance (1000W/m²), Gmes is the measured irradiance (W/m²), α is the temperature coefficient for current (A/°C), TSTC is the reference module temperature (°C), T is the measured module temperature (°C), VSTC is the module voltage (V), B is the temperature coefficient for voltage (V/°C), Imes is the measured current (A), Vmes is the measured voltage (V), Rsmes is the measured series resistance (Ω),Vt the thermal module voltage (V).

Equations (11) to (14), give the mathematical expression of each one of these parameter Iscref, Vocref, Imref, and Vmref.

$$I_{scref} = I_{scmes}\left(\frac{G_{STC}}{G_{mes}}\right) + \alpha\left(T - T_{STC}\right) \quad (11)$$

$$V_{ocref} = V_{ocmes} - \beta\left(T_{STC} - T\right) + V_t \ln\left(\frac{G_{STC}}{G_{mes}}\right) \quad (12)$$

$$I_{mref} = I_{mmes}\left(\frac{G_{STC}}{G_{mes}}\right) + \alpha\left(T - T_{STC}\right) \quad (13)$$

$$V_{mref} = V_{mmes} - \beta\left(T_{STC} - T\right) - R_{smes}\left(I_{mmes} - I_{m,ref}\right) + V_t \ln\left(\frac{G_{STC}}{G_{mes}}\right) \quad (14)$$

Introducing these parameters in the system equations (8), it can be solved with numeric or analytic methods and the four parameters I0ref, Aref, Rsref, and Rshref will be determined. We can determine the fifth parameter Iphref from (4).

IV. ANALYTIC METHOD

This method based on the approximations as in [3] allows the determination of the four parameters of the system equations (8).

The approximations used in [3] are given as follows

$$\exp\frac{V_{ocref}}{A_{ref}} >> \exp\frac{I_{scref}R_{sref}}{A_{ref}} \quad (15)$$

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$$R_{sref} \langle\langle R_{shref} \Rightarrow 1 + \frac{R_{sref}}{R_{shref}} \approx 1 \quad (16)$$

$$\frac{I_{0ref}}{A_{ref}} \exp\left(\frac{V_{ocref}}{A_{ref}}\right) >> \frac{1}{R_{shref}} \quad (17)$$

$$\frac{1}{A_{ref}} \exp \frac{I_{scref} R_{sref}}{A_{ref}} \approx 0 \quad (18)$$

Using these approximations we can found an analytic expression for each one of the five parameters of the characteristic equation (1) at reference conditions.

$$A_{ref} = \frac{V_{mref} + I_{mref} R_{S0} - V_{ocref}}{\left[\ln\left(I_{scref} - \frac{V_{mref}}{R_{shref}} - I_{mref}\right) - \ln\left(I_{scref} - \frac{V_{ocref}}{R_{shref}}\right) + \left(\frac{I_{mref}}{I_{scref} - \frac{V_{ocref}}{R_{shref}}}\right)\right]} \quad (19)$$

$$I_{0ref} = \left(I_{scref} - \frac{V_{ocref}}{R_{shref}}\right) \exp \frac{-V_{ocref}}{A_{ref}} \quad (20)$$

$$R_{sref} = R_{S0} - \frac{A_{ref}}{I_{0ref}} \exp \frac{-V_{ocref}}{A_{ref}} \quad (21)$$

$$R_{shref} = R_{sh0} \quad (22)$$

$$I_{phref} = I_{scref}\left(1 + \frac{R_{sref}}{R_{shref}}\right) + I_{0ref}\left\{\exp\left(\frac{I_{scref} R_{sref}}{A_{ref}}\right) - 1\right\} \quad (23)$$

V. Determination of the Five Parameters at Real Conditions

The five parameters which describe the I-V curve depend on cell temperature, on irradiance and on their reference values. This interdependence is given by the following equations as in [5].

A. The modified ideal factor

$$A = A_{ref}\left(\frac{T}{T_{ref}}\right) \quad (24)$$

B. The saturation current of the diode

$$I_0 = I_{0ref}\left[\frac{T}{T_{ref}}\right]^3 \exp\left[\frac{E_g N_s}{A_{ref}}\left(1 - \frac{T_{ref}}{T}\right)\right] \quad (25)$$

C. The photocurrent

$$I_{ph} = \frac{G}{G_{ref}}\left[I_{phref} + \alpha\left(T - T_{ref}\right)\right] \quad (26)$$

D. The series resistance Rs

$$R_s = R_{sref} \quad (27)$$

E.The shunt resistance Rsh

$$R_{sh} = R_{shref}\left(\frac{G_{ref}}{G}\right) \quad (28)$$

The following equations give Isc, Voc, Im, and Vm at any temperature and irradiance conditions [6]

$$I_{sc} = I_{scref}\left(\frac{G}{G_{ref}}\right) \quad (29)$$

$$V_{oc} = V_{ocref} - \beta\left(T_{ref} - T\right) \quad (30)$$

$$I_m = I_{mref}\left(\frac{G}{G_{ref}}\right) \quad (31)$$

$$V_m = V_{mref} - \beta\left(T_{ref} - T\right) \quad (32)$$

VI. Simulations

The current I is then evaluated using these parameters, variable voltage, irradiance and temperature.

In order to validate the simulation method presented above, a comparison is done between predicted characteristic and three cases of experimental measurement using a commercial module isophoton 106/12 composed of 36 mono-crystalline solar cells serially connected and tow parallel sides

A MATLAB script file is used to generate the simulation results.

Table I, II, II gives parameters values at reference and at real conditions for three experimental measures taken at different conditions of irradiance and temperature.

Case1: G=747W/m² and T=27.5°C

TABLE I
PARAMETERS VALUES AT REFERENCE AND AT REAL CONDITIONS

	Parameters At Reference Conditions	Parameters At Real Conditions
I_{ph} (A)	6.7119	5.0169
I_0 (A)	2.023^{-007}	1.5495^{-007}
A(V)	1.1587	1.1684
R_s(Ω)	0.1523	0.1523

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$R_{sh}(\Omega)$	110	147.2557
I_{sc} (A)	6.7026	5.0111
V_{oc} V)	20.0326	19.3167
I_m(A)	5.9797	4.4668
V_m (V)	16.3074	15.9324

CASE2: G=806 W/m² and T=27.1°C.

TABLE II
PARAMETERS VALUES AT REFERENCE AND AT REAL CONDITIONS

	Parameters At Reference Conditions	Parameters At Real Conditions
I_{ph} (A)	6.6797	5.3867
I_0 (A)	6.1769-007	5.0253-007
A(V)	1.2489	1.2577
$R_s(\Omega)$	0.1375	0.1375
$R_{sh}(\Omega)$	110	136.4764
I_{sc} (A)	6.6714	5.3771
V_{oc}(V)	20.1906	19.6043
I_m(A)	5.9394	4.7871
V_m (V)	16.3653	16.0503

Case3: G=762 W/m² and T=26.2°C.

TABLE III
PARAMETERS VALUES AT REFERENCE AND AT REAL CONDITIONS

	Parameters At Reference Conditions	Parameters At Real Conditions
I_{ph} (A)	6.7043	5.1102
I_0 (A)	1.4058^{-007}	1.2358^{-007}
A(V)	1.1516	1.1563
$R_s(\Omega)$	0.1531	0.1531
$R_{sh}(\Omega)$	110	144.3570
I_{sc} (A)	6.6950	5.1036
V_{oc}(V)	20.3270	19.8327
I_m(A)	5.9600	4.5416
V_m (V)	16.6370	16.4570

As be seen in table I, II, III, the five parameters at reference and at real conditions have almost the same values in the three different cases, which prove that the analytic method used for the extraction of these parameters gives good results.

Figure 2, 4, 6 shows the comparison between predicted characteristic and the experimental one fore the three studied cases.

Figure 3, 5, 7 shows the deviation between the predicted and experimental characteristic.

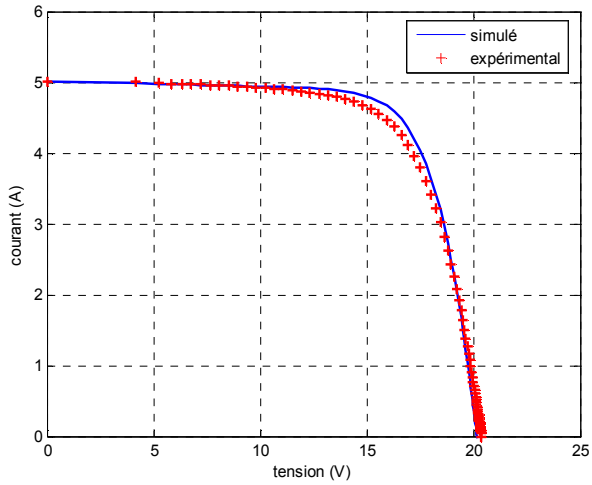

Fig. 2 Simulated and experimental characteristic for G=747W/m2 and T=27.5°C.

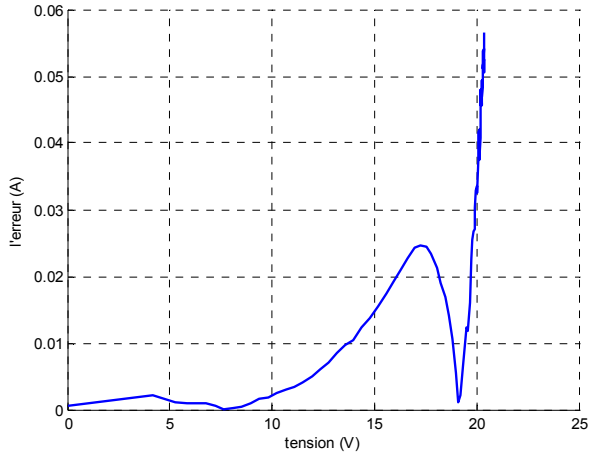

Fig. 3 Deviation between simulated and experimental characteristic for G=747W/m2 and T=27.5°C.

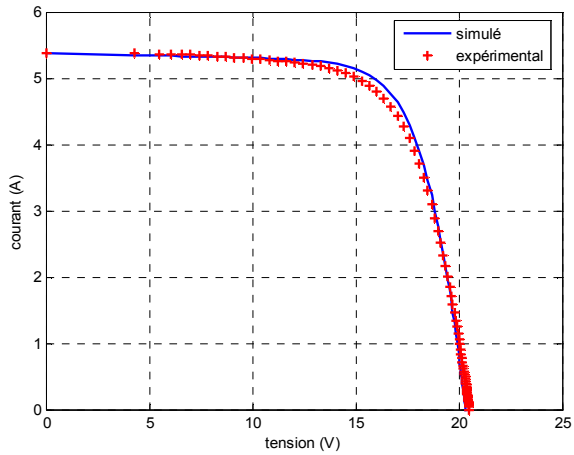

Fig. 4 Simulated and experimental characteristic for G=806 W/m2 and T=27.1°C.

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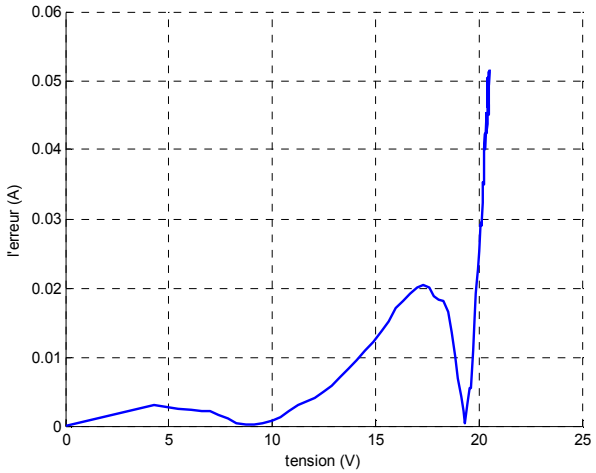

Fig. 5 Deviation between simulated and experimental characteristic for G=806 W/m2 and T=27.1°C.

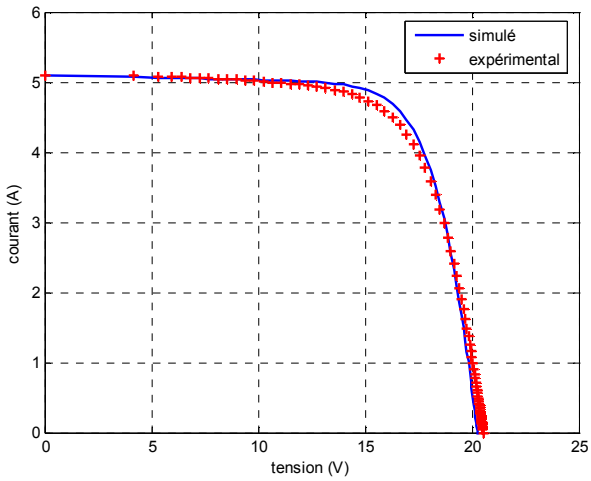

Fig. 6 Simulated and experimental characteristic for G=762 W/m² and T=26.2°C.

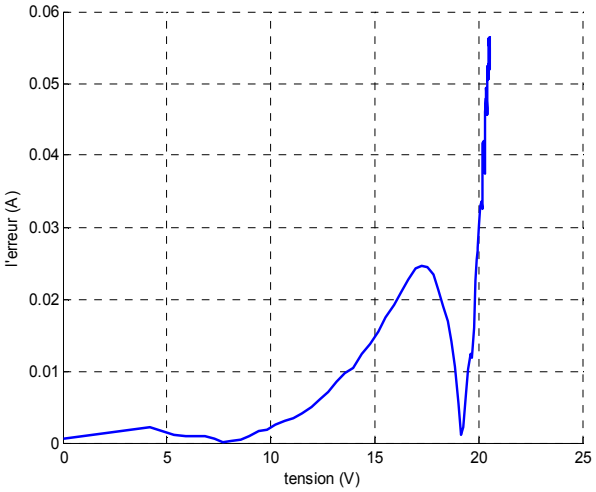

Fig. 7 Deviation between simulated and experimental characteristic for G=762 W/m² and T=26.2°C.

As can be seen in figures 2,4, 6, simulation results give a good approach to real I-V characterstic.

VII. Error Calculation

We propose to calculate the tow following indexes of error

A. The deviation between predicted and experimental characteristic using the RMSE error (Root Mean Square Error) given by the following relation [5]:

$$RMSE(A) = \sqrt{\frac{\sum_{i=1}^{N}\left(I_{ical} - I_{i\exp}\right)^2}{N}} \tag{33}$$

Where Iical is the calculated or predicted current and Iiexp is the experimental current, N the number of measured points.

The RMSE in percent is also given by:

$$RMSE\,(\%) = 100\,\frac{\sqrt{\frac{\sum_{i=1}^{N}\left(I_{ical} - I_{i\exp}\right)^2}{N}}}{I_{sc\exp}} \tag{34}$$

B. The mean relative error of calculated parameters Iscref, Vocref, Imref, and Vmref given by the following expression.

$$E_X = 100\,\frac{X_{cal} - X_{\exp}}{X_{\exp}}\,(\%) \tag{35}$$

Where X = [Isc, Voc, Im, Vm].

Table IV gives the values of relative error, RMSE (A) and RMSE (%).

TABLE IV
RELATIVE ERROR AND RMSE

Error	Values		
	Case 1	Case 2	Case 3
E_{Isc} (%)	0.3614	0.4236	0.2441
E_{Voc} (%)	-5.3101	-4.3691	-3.2550
E_{Im} (%)	0.3537	0.4318	0.2490
E_{Vm} (%)	0.0779	-0.3089	2.2176
RMSE(A)	0.3584	0.3255	0.5012
RMSE(%)	7.1538	6.0505	9.8276

The deviation is almost negligibal at shurt circuit current point, and at maximum power point.in addition an error not exeeding 3.67% is noted at open circuit voltage point.

VIII. Dynamic Behaviour of Photovoltaic Module

For constant irradiance and temperature the I-V characteristic describes the static behaviour of the PV module but for variable irradiance and temperature during a day, the current I will be evaluated depending time. We introduce in the model of the photovoltaic generator a profile of temperature and irradiance during four days of March stored in the central acquisition (sampling period=1minute).

This simulation is done on 15 modules serially associated and tow parallel sides which gives a generator composed of 30 modules.

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Figure (7) shows the comparison between predicted and experimental current at maximum power point.

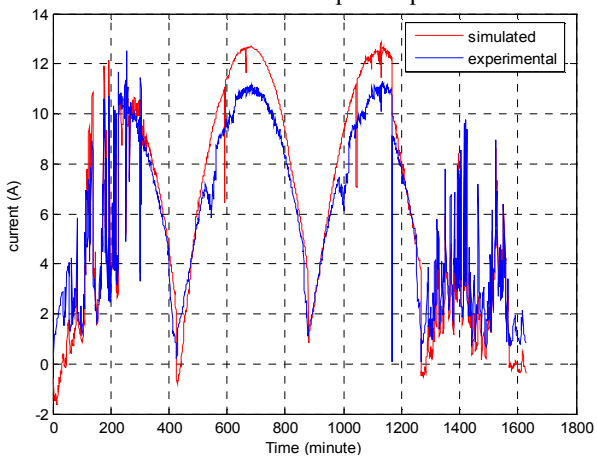

Fig. 8 Comparison between simulated and experimental current at maximum power point.

The maximum power generated by the photovoltaic generator is given by the following relation:

$$P_{\max} = I_m V_m \qquad 36$$

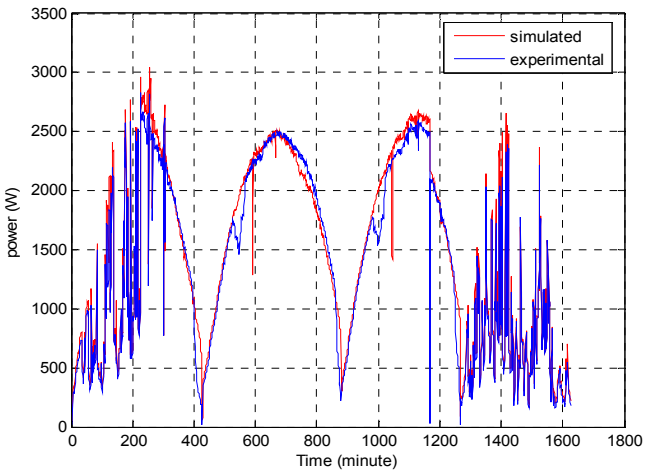

Fig. 9 Comparison between simulated and experimental maximum power.

Table V shows that the error between simulated measured current at maximum power point is reasonable, the same thing for the maximum power which gives an acceptable registered error relatively with the global provided power.

TABLE V
RELATIVE ERROR AND RMSE FOR CURRNT AND POWER.

Error	Current*(A)*	Power*(W)*
RMSE	*1.3299*	*152.5778*
RMSE (%)	*9.9556*	*5.1203*

IX. CONCLUSION

This paper indicates how the five parameters of the single diode model are determined and gives their values at reference conditions. These values are then used with in the model to calculate the five parameters at any other conditions. For the determination of the characteristic parameters such as the short circuit current, the open circuit voltage, the current and voltage at maximum power point, translation equations are used.

Comparison between predicted current-voltage curves and experimental data performed at photovoltaic laboratory of the center of development renewable energy, gives good results with acceptable errors. These results shown that the single one diode model can be an accurate tool for the prediction of energy production.

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